# Notes: Berman, Martin, and Mayer (QJE, 2012)

## How Do Different Exporters React to Exchange Rate Changes?

## Contents

# 1 Big picture

- **Question.** How do exporters with different performance levels react to exchange rate changes in prices, quantities, entry, and aggregate exports?
- **Core claim.** Exchange rate pass-through and export-volume responses are heterogeneous across firms: high-performance exporters absorb more exchange rate movements in markups and adjust quantities less.
- **Empirical setting.** The paper uses French firm-level customs data matched to balance-sheet information for 1995–2005. The data report export values and quantities by firm, product, destination, and year, which allows unit values to proxy FOB export prices.
- **Main price result.** Following a depreciation, high-performance firms increase export prices in euros more than low-performance firms. This is interpreted as stronger pricing-to-market by the best exporters.
- **Main quantity result.** High-performance firms' export volumes respond less to depreciations because more of the exchange-rate change is absorbed in markups rather than passed through to foreign-currency prices.
- **Extensive margin result.** A depreciation increases the probability of exporting to a destination, both through entry into destinations and through higher continuation probabilities.
- **Aggregate implication.** Aggregate exports are concentrated among the highest-performance firms, precisely the firms with the lowest volume elasticity. This helps explain why aggregate exports respond weakly to exchange rate movements.
- **Theory connection.** The results are consistent with models in which the perceived demand elasticity falls with firm performance, such as variable-markup models, quality models, and models with local distribution costs.
- **Economic takeaway.** Representative-firm export elasticities miss an important composition effect: the dominant exporters are the firms that most insulate quantities from exchange-rate movements.

# 2 Incremental contribution of the paper

## 2.1 What the paper adds relative to exchange-rate pass-through studies

- Earlier exchange-rate pass-through work mostly studies import prices, consumer prices, or aggregate trade flows. Those data cannot identify how individual exporters with different productivity levels adjust markups and quantities.
- This paper uses firm-destination-product export values and quantities, allowing the authors to observe unit values at the exporter-destination level and therefore infer FOB price adjustments by individual firms.
- The paper shows that the average pass-through coefficient hides large firm-level heterogeneity. High-performance exporters price much more to market, while low-performance exporters pass exchange-rate changes more strongly into prices faced by foreign consumers.

**Interpretation:** The incremental contribution is not simply documenting incomplete pass-through. The paper explains why pass-through is incomplete at the aggregate level: the most important exporters are the ones that absorb exchange-rate shocks in markups.

## 2.2 What the paper adds relative to heterogeneous-firm trade models

- Standard Melitz-style models generate firm heterogeneity in export participation and size, but with constant-elasticity demand they do not naturally generate heterogeneous pricing-to-market.

- The paper connects the empirical pattern to models with variable markups, non-CES demand, local distribution costs, and quality differentiation.
- It provides direct micro evidence for the prediction that exchange-rate elasticities of prices and quantities should vary systematically with firm performance.

**Interpretation:** The paper turns a theoretical mechanism in heterogeneous-firm models into an empirical object that can be measured with customs data.

### 2.3 What the paper adds relative to aggregate export elasticity puzzles

- Aggregate exports often appear surprisingly insensitive to real exchange rates.
- The paper argues that concentration matters: if large high-productivity exporters have low volume elasticities, aggregate exports can be insensitive even when smaller firms react strongly.
- Sector-level and calibration exercises show that heterogeneous pricing-to-market can reduce aggregate export elasticities and partly reconcile micro and macro evidence.

**Interpretation:** The paper's aggregate contribution is a composition channel: the elasticity of total exports is not the average firm elasticity; it is a weighted average dominated by high-performance firms.

## 3 Data and measurement

### 3.1 French customs data

- French customs data report the value and volume of exports by firm, product, destination, and year.
- Products are defined at the eight-digit Combined Nomenclature level.
- Unit values are computed as export value divided by export quantity and are used as proxies for FOB exporter prices.
- The empirical sample covers 1995–2005.
- The analysis excludes Eurozone destinations to retain meaningful exchange-rate variation.

**Interpretation:** Using FOB export unit values is important because import-price or consumer-price data include shipping, distribution, and local retail margins that may themselves react to exchange rates.

### 3.2 Balance-sheet data and firm performance

- Firm balance-sheet variables come from the BRN database, which includes sales, value added, employment, capital, and sector of activity.
- The authors match customs data to BRN, which covers firms accounting for most French exports.
- Firm performance is measured primarily by total factor productivity (TFP), estimated sector by sector using the Olley-Pakes method.
- The main alternative performance measure is apparent labor productivity, value added per worker.
- Additional performance proxies include product rank in the firm export portfolio and the number of destinations served by the firm.

**Interpretation:** Because performance is measured before the exchange-rate response, the paper can ask whether ex ante better firms react differently to exchange-rate changes.

### 3.3 Exchange rates and macro controls

- The key macro variable is the destination-specific real exchange rate, denoted $RER_{it}$.
- The exchange rate is defined as the price of the domestic currency, the euro, in foreign currency units. Thus an increase in $RER$ is a depreciation of the euro.
- Destination GDP and importer price indexes are included as controls in volume regressions.
- The importer price index captures destination-specific price-level variation that affects foreign demand.

**Interpretation:** The focus is on destination-year exchange-rate changes. Firm-destination fixed effects absorb permanent differences in firm-market relationships.

### 3.4 Samples used in the paper

The paper constructs several samples to address multi-product and composition issues:

- **Single product:** firm-destination pairs with a single exported product over the period.
- **Main product by value:** the product with the largest average export value within the firm.
- **Main product by destinations:** the product exported by the firm to the largest number of destinations.
- **Stable mix:** firm-destination observations with stable product composition.
- **Single NC4:** product aggregation at the four-digit level.
- **Firm level:** exports aggregated across products by firm-destination-year.
- **Firm-product level:** a larger panel allowing multiple products per firm.

**Interpretation:** The multiple samples show that results are not mechanically driven by changes in product composition or by unit-value aggregation.

## 4 Theory and testable predictions

### 4.1 Mechanism: variable demand elasticity

High-performance firms tend to have larger market shares, higher markups, or higher quality. In models with variable demand elasticity, these firms perceive a lower demand elasticity. When the home currency depreciates, they can raise markups rather than fully pass the depreciation into lower foreign-currency prices.

**Interpretation:** The central mechanism is endogenous markups. Exchange-rate changes alter the firm's optimal markup, and the markup adjustment is larger for high-performance firms.

### 4.2 Testable Prediction 1: export prices

Let $p_i(\varphi)$ be the exporter price in home currency for a firm with performance $\varphi$ selling to destination $i$, and let $q_i$ be the real exchange rate. Define the exporter-price elasticity:

$$e_{p_i}(\varphi) = \frac{\partial p_i(\varphi)}{\partial q_i}\frac{q_i}{p_i(\varphi)}. \tag{1}$$

**Prediction 1:**

$$\frac{\partial e_{p_i}(\varphi)}{\partial \varphi} > 0. \tag{2}$$

**Interpretation:** High-performance firms should increase home-currency export prices more after a depreciation. Equivalently, they absorb more of the exchange-rate change in their markup.

### 4.3 Testable Prediction 2: export volumes

Let $x_i(\varphi)$ be export volume. Define the volume elasticity:

$$e_{x_i}(\varphi) = \frac{\partial x_i(\varphi)}{\partial q_i} \frac{q_i}{x_i(\varphi)}. \tag{3}$$

**Prediction 2:**

$$\frac{\partial e_{x_i}(\varphi)}{\partial \varphi} < 0. \tag{4}$$

**Interpretation:** If high-performance firms raise markups more, their foreign-currency prices fall less after a depreciation, so their export quantities expand less.

### 4.4 Distribution-cost channel

In a model with local distribution costs, consumer prices combine an exporter price component and a local distribution component. Since local distribution costs are paid in destination currency, a depreciation reduces the exporter-price share of the consumer price. This lowers the elasticity of demand perceived by the exporter, especially for high-productivity firms.

**Interpretation:** The distribution-cost model predicts that pricing-to-market and heterogeneous responses should be stronger in destinations or sectors where distribution costs are larger.

## 5 Empirical specifications

### 5.1 Unit-value regression

The baseline price regression is:

$$\ln(UV_{jit}) = \alpha_p \ln(\varphi_{j,t-1}) + \beta_p \ln(RER_{it}) + \gamma_p \left[\ln(\varphi_{j,t-1}) \times \ln(RER_{it})\right] + \psi_t + \mu_{ji} + \varepsilon_{jit}. \tag{5}$$

- $UV_{jit}$ is the unit value of exports by firm $j$ to destination $i$ at time $t$.
- $\varphi_{j,t-1}$ is lagged firm performance, usually TFP or labor productivity.
- $RER_{it}$ is the real exchange rate for destination $i$.
- $\mu_{ji}$ are firm-destination fixed effects.
- $\psi_t$ are year effects.

**Interpretation:** The coefficient $\gamma_p$ measures whether more productive firms change their prices differently in response to real exchange-rate movements. Prediction 1 implies $\gamma_p > 0$.

### 5.2 Export-volume regression

The baseline volume regression is:

$$\begin{aligned} \ln(x_{jit}) = {} & \alpha_x \ln(\varphi_{j,t-1}) + \beta_x \ln(RER_{it}) + \gamma_x \left[\ln(\varphi_{j,t-1}) \times \ln(RER_{it})\right] \\ & + \delta_1 \ln(GDP_{it}) + \delta_2 \ln(ImporterPriceIndex_{it}) + \psi_t + \mu_{ji} + \varepsilon_{jit}. \end{aligned} \tag{6}$$

**Interpretation:** Prediction 2 implies $\gamma_x < 0$: more productive firms should increase export quantities less when the home currency depreciates.

### 5.3 Extensive-margin regression

The paper estimates export participation as:

$$Export_{jit} = 1\{x_{jit} > 0\}, \tag{7}$$

using logit and linear probability models. The main regressors are lagged TFP, the real exchange rate, destination GDP, and the importer price index.

**Interpretation:** The extensive-margin exercise asks whether depreciations make it more likely that a firm exports to a destination, including entry into new firm-destination markets and continuation in existing markets.

### 5.4 Sector-level aggregation

At the sector-destination level, the paper relates aggregate export values or volumes to exchange rates and sectoral measures of heterogeneity:

$$\ln(X_{sit}) = \beta \ln(RER_{it}) + \lambda H_{s,t-1} + \rho \left[H_{s,t-1} \times \ln(RER_{it})\right] + Controls + FE + \varepsilon_{sit}. \tag{8}$$

**Interpretation:** A negative interaction between sector heterogeneity and the exchange-rate elasticity means sectors dominated by very large or high-productivity firms respond less to exchange-rate movements.

## 6 Identification and empirical design

### 6.1 Main identifying variation

- The design uses destination-year variation in real exchange rates and firm-level variation in performance.
- Firm-destination fixed effects absorb persistent differences in a firm's relationship with a destination market.
- Year effects absorb aggregate shocks common to all firms and destinations.
- The interaction between firm performance and destination-specific exchange rates identifies heterogeneous pricing-to-market.

**Interpretation:** Identification comes from whether higher- and lower-performance firms selling to the same destination-year environment adjust differently when the exchange rate changes.

### 6.2 Why firm-destination fixed effects matter

Firm-destination fixed effects absorb:

- permanent differences in distance, language, and market access;
- persistent quality differences in firm-destination matches;
- stable firm-market demand relationships;
- persistent firm-destination price levels.

**Interpretation:** The fixed effects make the test a within-firm-destination comparison over time, rather than a cross-sectional comparison of different exporters.

## 6.3 Threats and robustness

- Unit values may reflect quality changes or product mix, so the paper uses several samples that hold product composition more fixed.
- Productivity measures may be noisy, so the paper uses both TFP and labor productivity.
- Results may reflect rank or product heterogeneity, so the paper controls for product rank and uses firm-product specifications.
- Local distribution costs and intermediate-good channels may confound the mechanism, so the paper directly tests these channels.

**Interpretation:** The robustness exercises show that the heterogeneous exchange-rate response is not a simple artifact of sample definition, productivity measure, or product aggregation.

# 7 Tables and figures

## 7.1 Tables I and II: sample and descriptive statistics

**Read:**

- Table I reports how each sample represents French exports. Depending on the sample, the coverage ranges from narrow single-product firm-destination observations to a very large firm-product-level sample.
- Table II reports descriptive statistics for firm size, labor productivity, number of destinations, and growth in unit values and export volumes.
- The average firm-product-level observation is much larger than the median, showing strong skewness in exporter size.
- Median growth in both unit values and export volumes is positive in most samples, while mean growth can be affected by outliers.

**Economic message:** The data are large enough to study heterogeneous price and quantity reactions at a granular firm-destination-product level, while still representing a large share of French exports.

TABLE I

SAMPLE REPRESENTATIVENESS

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| le | Single product | Main Product (val.) | Main Product (dest.) | Stable mix | Single NC4 | Firm level | Firm-product level |
| ervations | 2233747 | 1986364 | 2875547 | 2335437 | 2650444 | 4193368 | 12489069 |
| ns | 263509 | 284169 | 284169 | 264040 | 268589 | 284169 | 284169 |
| nch export value[a] | 12% | 50% | 33% | 14% | 18% | 100% | 100% |
| nch export val. firms in sample[b] | 95% | 100% | 100% | 96% | 96% | 100% | 100% |

te. Authors' computation from French customs data, 1995–2005. a. Share of total French exports (minus Eurozone) present in this subsample. b. Share of total French exports Eurozone) by firms present in this subsample.

*Table I: sample representativeness*

TABLE II

DESCRIPTIVE STATISTICS

| Variable | Obs. | Mean | Median | Std.Dev. |
|---|---|---|---|---|
| Single prod. obs. | | | | |
| # employees | 949,549 | 263.3 | 63.0 | 1246.1 |
| value added/worker | 949,549 | 58.7 | 49.9 | 34.0 |
| # destinations | 949,549 | 21.7 | 15.0 | 21.4 |
| growth in unit val. (%) | 398,928 | 1.6 | 0.7 | 49.5 |
| growth in export vol. (%) | 398,928 | −1.2 | 0.0 | 91.1 |
| Main product (value) | | | | |
| # employees | 973,984 | 475.3 | 106.0 | 2173.9 |
| value added/worker | 973,984 | 62.4 | 53.1 | 36.1 |
| # destinations | 973,984 | 30.0 | 23.0 | 26.6 |
| growth in unit val. (%) | 572,732 | 0.7 | 0.6 | 38.3 |
| growth in export vol. (%) | 572,732 | 3.0 | 2.9 | 84.8 |
| Main product (dest.) | | | | |
| # employees | 1,129,958 | 471.6 | 102.0 | 2086.2 |
| value added/worker | 1,129,958 | 61.4 | 52.46 | 35.1 |
| # destinations | 1,129,958 | 29.1 | 22.0 | 26.4 |
| growth in unit val. (%) | 659,307 | 0.8 | 0.5 | 43.5 |
| growth in export vol. (%) | 659,307 | 2.7 | 2.5 | 87.7 |
| Stable mix | | | | |
| # employees | 1,005,869 | 272.5 | 66.0 | 1246.5 |
| value added/worker | 1,005,869 | 59.1 | 50.2 | 34.3 |
| # destinations | 1,005,869 | 22.3 | 16.0 | 21.8 |
| growth in unit val. (%) | 435,111 | 1.7 | 0.7 | 48.8 |
| growth in export vol. (%) | 435,111 | −2.6 | 0.0 | 90.2 |
| Single prod. obs. (NC4) | | | | |
| # employees | 1,171,626 | 294.2 | 72.0 | 1309.4 |
| value added/worker | 1,171,626 | 59.0 | 50.2 | 34.1 |
| # destinations | 1,171,626 | 23.2 | 16.0 | 22.3 |
| growth in unit val. (%) | 466,784 | 1.7 | 0.8 | 47.8 |
| growth in export vol. (%) | 466,784 | −1.5 | 0.0 | 90.5 |
| Firm-level | | | | |
| # employees | 1,922,646 | 445.9 | 89.0 | 2039.4 |
| value added/worker | 1,922,646 | 59.7 | 51.3 | 33.7 |
| # destinations | 1,922,646 | 26.3 | 24.8 | 19.0 |
| # product by dest. | 1,922,646 | 3.1 | 2.0 | 6.0 |
| growth in unit val. (%) | 1,180,233 | 1.6 | 0.8 | 48.9 |
| growth in export vol. (%) | 1,180,233 | −0.1 | 1.1 | 89.9 |

*Table II: descriptive statistics, main panel*

TABLE II

(CONTINUED)

| Variable | Obs. | Mean | Median | Std.Dev. |
|---|---|---|---|---|
| Firm-product level | | | | |
| # employees | 5,287,169 | 1497.9 | 176.0 | 5461.4 |
| value added/worker | 5,287,169 | 64.5 | 55.7 | 37.3 |
| # destinations / product | 5,287,169 | 12.0 | 4.0 | 17.6 |
| growth in unit val. (%) | 2,558,069 | 0.5 | 0.4 | 55.8 |
| growth in export vol. (%) | 2,558,069 | 2.1 | 1.3 | 97.8 |

*Note.* Value added per worker in thousands of euros. Authors' computations from BRN/French Custom data. Non–Eurozone destinations, positive exports observations. We use the growth measure initially

*Table II continued*

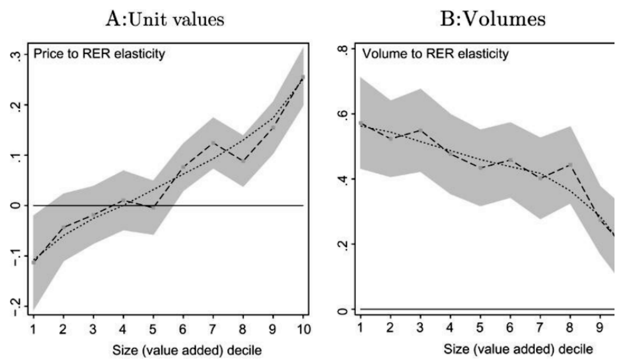


*Figure I: responses by size decile*

## 7.2 Table III: baseline results

**Read:**

- The upper panel uses log unit values as the dependent variable.
- The coefficient on $\ln RER$ is positive, so a euro depreciation raises euro export prices.
- The interaction $\ln TFP \times \ln RER$ is positive and significant across samples, supporting Prediction 1.
- The lower panel uses log export volume as the dependent variable.
- The interaction $\ln TFP \times \ln RER$ is negative and significant across samples, supporting Prediction 2.
- Quantification rows show that exchange-rate elasticities differ substantially between low- and high-productivity firms.

**Economic message:** High-productivity exporters absorb more of the depreciation in markups and expand quantities less. This is the core firm-level evidence.



TABLE III

BASELINE RESULTS

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| Sample | Single product | Main product (val.) | Main product (dest.) | Stable mix | Single NC4 | Firm level | Firm-product level |
| # observations | 355996 | 429022 | 486403 | 364672 | 489079 | 858271 | 2289051 |
| Dep. var. | ln unit value | | | | | | |
| | Coefficients | | | | | | |
| ln TFP$_{t-1}$ | 0.012$^{a}$ | 0.018$^{a}$ | 0.006$^{b}$ | 0.014$^{a}$ | 0.012$^{a}$ | 0.010$^{a}$ | 0.010$^{a}$ |
| | (0.004) | (0.003) | (0.003) | (0.004) | (0.003) | (0.002) | (0.002) |
| ln RER | 0.084$^{a}$ | 0.135$^{a}$ | 0.108$^{a}$ | 0.097$^{a}$ | 0.078$^{a}$ | 0.052$^{a}$ | 0.124$^{a}$ |
| | (0.019) | (0.015) | (0.016) | (0.018) | (0.016) | (0.017) | (0.020) |
| ln TFP$_{t-1}\times$ ln RER | 0.047$^{a}$ | 0.059$^{a}$ | 0.055$^{a}$ | 0.042$^{a}$ | 0.040$^{a}$ | 0.024$^{a}$ | 0.023$^{a}$ |
| | (0.015) | (0.009) | (0.009) | (0.014) | (0.013) | (0.009) | (0.008) |
| rank product | | | | | | | –0.003$^{a}$ |
| | | | | | | | (0.000) |
| rank product $\times$ ln RER | | | | | | | –0.003$^{a}$ |
| | | | | | | | (0.001) |
| | Quantification: change in the effect of RER (%), for | | | | | | |
| mean TFP$\rightarrow$ mean + s.d TFP | 8.4 $\rightarrow$ 13.4 | 13.5 $\rightarrow$ 19.5 | 10.8 $\rightarrow$ 16.4 | 9.7 $\rightarrow$ 14.1 | 7.8 $\rightarrow$ 12.2 | 5.2 $\rightarrow$ 7.9 | 12.4 $\rightarrow$ 15.2 |
| 1st$\rightarrow$ 5th product | | | | | | | 12.4 $\rightarrow$ 11.0 |
| 1st$\rightarrow$ 10th product | | | | | | | 12.4 $\rightarrow$ 9.3 |
| Dep. var. | ln volume | | | | | | |
| | Coefficients | | | | | | |
| ln TFP$_{t-1}$ | 0.082$^{a}$ | 0.125$^{a}$ | 0.115$^{a}$ | 0.089$^{a}$ | 0.097$^{a}$ | 0.104$^{a}$ | 0.076$^{a}$ |
| | (0.008) | (0.007) | (0.007) | (0.009) | (0.007) | (0.006) | (0.006) |

*Table III, unit values*

TABLE III

(CONTINUED)

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| Sample | Single product | Main product (val.) | Main product (dest.) | Stable mix | Single NC4 | Firm level | Firm-product level |
| # observations | 355996 | 429022 | 486403 | 364672 | 489079 | 858271 | 2289051 |
| ln RER | 0.399$^{a}$ | 0.542$^{a}$ | 0.560$^{a}$ | 0.419$^{a}$ | 0.498$^{a}$ | 0.704$^{a}$ | 0.481$^{a}$ |
| | (0.044) | (0.059) | (0.057) | (0.054) | (0.048) | (0.070) | (0.055) |
| ln TFP$_{t-1}\times$ ln RER | –0.105$^{a}$ | –0.074$^{b}$ | –0.075$^{b}$ | –0.052 | –0.091$^{a}$ | –0.006 | 0.022 |
| | (0.035) | (0.029) | (0.029) | (0.034) | (0.033) | (0.027) | (0.033) |
| rank product | | | | | | | –0.060$^{a}$ |
| | | | | | | | (0.002) |
| rank product $\times$ ln RER | | | | | | | 0.015$^{b}$ |
| | | | | | | | (0.007) |
| ln GDP | 0.628$^{a}$ | 0.942$^{a}$ | 0.941$^{a}$ | 0.725$^{a}$ | 0.744$^{a}$ | 0.984$^{a}$ | 0.849$^{a}$ |
| | (0.051) | (0.071) | (0.063) | (0.055) | (0.055) | (0.073) | (0.057) |
| ln importer price index | 0.054$^{a}$ | 0.088$^{a}$ | 0.085$^{a}$ | 0.064$^{a}$ | 0.056$^{a}$ | 0.081$^{a}$ | 0.072$^{a}$ |
| | (0.012) | (0.016) | (0.015) | (0.014) | (0.011) | (0.016) | (0.013) |
| | Quantification: change in the effect of RER (%), for | | | | | | |
| mean TFP $\rightarrow$ mean + s.d TFP | 39.9$\rightarrow$28.5 | 54.2$\rightarrow$46.6 | 56.0$\rightarrow$48.4 | 41.9$\rightarrow$36.5 | 49.8$\rightarrow$40.0 | 70.4$\rightarrow$69.8 | 48.1$\rightarrow$50.8 |
| 1st $\rightarrow$ 5th product | | | | | | | 48.1$\rightarrow$54.3 |
| 1st $\rightarrow$ 10th product | | | | | | | 48.1$\rightarrow$61.9 |

*Note.* Robust standard errors clustered by destination-year in parentheses with [a], [b], and [c], respectively, denoting significance at 1%, 5%, and 10%. Columns (1) to (6) include firm-destination fixed effects and year dummies. Column (7) has firm-destination-product fixed effects together with year dummies. TFP is demeaned, and the rank product variables are computed by firm-destination, and normalized such that the core product has rank 0.



*Table III, volumes*

## 7.3 Tables IV and V: alternative performance measures

**Read:**

- Table IV replaces TFP with labor productivity and finds the same pattern: more productive firms increase prices more and volumes less after a depreciation.
- Table V uses performance bins or percentiles rather than a linear productivity interaction.
- The coefficients become stronger in higher performance groups, consistent with a monotone but nonlinear effect.
- These tables show that the baseline results are not driven by one specific productivity estimator.

**Economic message:** The result is a robust performance gradient. It appears whether performance is measured by TFP, labor productivity, or percentile bins.

TABLE IV
ROBUSTNESS: LABOR PRODUCTIVITY

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Sample | Single product | Main product (val.) | Main product (dest.) | Stable mix | Single NC4 | Firm level |
| # observations | 355996 | 429022 | 486403 | 364672 | 489079 | 858270 |
| Dep. var. | ln unit value | | | | | |
| | Coefficients | | | | | |
| ln Labor Prod.$_{t-1}$ | 0.016[a] | 0.027[a] | 0.013[a] | 0.015[a] | 0.013[a] | 0.013[a] |
| | (0.005) | (0.003) | (0.003) | (0.004) | (0.004) | (0.002) |
| ln RER | 0.073[a] | 0.140[a] | 0.116[a] | 0.104[a] | 0.087[a] | 0.064[a] |
| | (0.019) | (0.015) | (0.016) | (0.018) | (0.016) | (0.016) |
| ln Labor Prod.$_{t-1}$ × ln RER | 0.099[a] | 0.095[a] | 0.100[a] | 0.103[a] | 0.087[a] | 0.074[a] |
| | (0.016) | (0.012) | (0.013) | (0.015) | (0.014) | (0.010) |
| | Quantification: change in the effect of RER (%), for | | | | | |
| mean prod. → mean + s.d prod. | 7.3 → 30.8 | 14.0 → 28.9 | 11.6 → 27.2 | 10.4 → 33.0 | 8.7 → 29.5 | 6.4 → 24.4 |

*Table IV: labor productivity, unit values*

TABLE IV
(CONTINUED)

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Sample | Single product | Main product (val.) | Main product (dest.) | Stable mix | Single NC4 | Firm level |
| # observations | 355996 | 429022 | 486403 | 364672 | 489079 | 858270 |
| Dep. var. | ln volume | | | | | |
| | Coefficients | | | | | |
| ln Labor Prod.$_{t-1}$ | 0.078[a] | 0.142[a] | 0.121[a] | 0.089[a] | 0.099[a] | 0.126[a] |
| | (0.009) | (0.008) | (0.008) | (0.009) | (0.008) | (0.007) |
| ln RER | 0.410[a] | 0.531[a] | 0.555[a] | 0.417[a] | 0.495[a] | 0.686[a] |
| | (0.044) | (0.059) | (0.057) | (0.053) | (0.048) | (0.069) |
| ln Labor Prod.$_{t-1}$ × ln RER | −0.118[a] | −0.144[a] | −0.113[a] | −0.078[b] | −0.106[a] | −0.086[a] |
| | (0.038) | (0.032) | (0.031) | (0.037) | (0.035) | (0.030) |
| ln GDP | 0.629[a] | 0.943[a] | 0.941[a] | 0.726[a] | 0.745[a] | 0.985[a] |
| | (0.051) | (0.070) | (0.063) | (0.055) | (0.055) | (0.073) |
| ln importer price index | 0.054[a] | 0.088[a] | 0.085[a] | 0.064[a] | 0.056[a] | 0.081[a] |
| | (0.012) | (0.016) | (0.015) | (0.014) | (0.011) | (0.016) |
| | Quantification: change in the effect of RER (%), for | | | | | |
| mean prod. → mean + s.d prod. | 41.0 → 12.9 | 53.1 → 30.4 | 55.5 → 38.0 | 41.7 → 24.7 | 49.5 → 24.3 | 68.6 → 47.5 |

*Note.* Robust standard errors clustered by destination-year in parentheses with a, b, and c, respectively, denoting significance at 1%, 5%, and 10%. Columns (1) to (6) include firm-destination fixed effects and year dummies. Labor productivity is demeaned.

*Table IV: labor productivity, volumes*

TABLE V
ROBUSTNESS: PERCENTILES

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dep. var | | ln unit value | | | ln volume | |
| # of bins | 2 | 5 | 10 | 2 | 5 | 10 |
| ln RER | 0.068[a] | 0.070[a] | 0.091[a] | 0.436[a] | 0.484[a] | 0.471[a] |
| | (0.020) | (0.025) | (0.031) | (0.048) | (0.056) | (0.065) |
| Top 50% TFP × ln RER | 0.035[b] | | | −0.077[b] | | |
| | (0.018) | | | (0.039) | | |
| Top 20% TFP × ln RER | | 0.074[a] | | | −0.188[a] | |
| | | (0.026) | | | (0.064) | |
| Top 10% TFP × ln RER | | | 0.087[b] | | | −0.166[c] |
| | | | (0.035) | | | (0.085) |
| ln GDP | | | | 0.628[a] | 0.628[a] | 0.627[a] |
| | | | | (0.051) | (0.051) | (0.051) |
| ln importer price index | | | | 0.054[a] | 0.054[a] | 0.054[a] |
| | | | | (0.012) | (0.012) | (0.012) |
| | Quantification: change in the effect of RER (%), for | | | | | |
| Benchmark group → Top group | 6.8 → 10.3 | 7.0 → 14.4 | 9.1 → 17.8 | 43.6 → 35.9 | 48.4 → 29.6 | 47.1 → 30.5 |

*Table V: percentiles, unit values*

TABLE V
(CONTINUED)

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| Dep. var | | ln unit value | | | ln volume | |
| # of bins | 2 | 5 | 10 | 2 | 5 | 10 |
| ln RER | 0.017 | 0.038 | 0.059[c] | 0.462[a] | 0.467[a] | 0.438[a] |
| | (0.022) | (0.026) | (0.032) | (0.050) | (0.057) | (0.069) |
| Top 50% Labor Prod. × ln RER | 0.120[a] | | | −0.116[a] | | |
| | (0.019) | | | (0.045) | | |
| Top 20% Labor Prod. × ln RER | | 0.147[a] | | | −0.188[a] | |
| | | (0.028) | | | (0.069) | |
| Top 10% Labor Prod. × ln RER | | | 0.169[a] | | | −0.190[b] |
| | | | (0.037) | | | (0.081) |
| ln GDP | | | | 0.629[a] | 0.629[a] | 0.629[a] |
| | | | | (0.051) | (0.051) | (0.051) |
| ln importer price index | | | | 0.054[a] | 0.054[a] | 0.054[a] |
| | | | | (0.012) | (0.012) | (0.012) |
| | Quantification: change in the effect of RER (%), for | | | | | |
| Benchmark group → Top group | 1.7 → 13.7 | 3.8 → 18.5 | 5.9 → 22.8 | 46.2 → 34.6 | 46.7 → 27.9 | 43.8 → 24.9 |

*Note.* Number of observations is 355,996 in all regressions. Robust standard errors clustered by destination-year in parentheses with [a, b], and [c], respectively, denoting significance at 1%, 5%, and 10%. Columns (1) to (6) include firm-destination fixed effects and year dummies. Percentiles computed by year. All regressions include TFP/Labor Prod. bins dummies as well as the full set of bins interactions with ln RER. The benchmark group is the bottom TFP/Labor Prod. bin in each column.

*Table V: percentiles, volumes*

## 7.4 Figures I and II: nonparametric heterogeneity

**Read:**

- Figure I plots price and volume exchange-rate elasticities across value-added size deciles.
- Price elasticities rise strongly with size; volume elasticities fall with size.
- Figure II repeats the exercise by bins of the number of destinations served by the firm.
- Firms serving many destinations display stronger pricing-to-market and lower quantity responses.

**Economic message:** The graphical evidence shows that the interaction terms are not hiding a fragile linear fit. The heterogeneity is visible across the distribution of firm size and exporter scope.

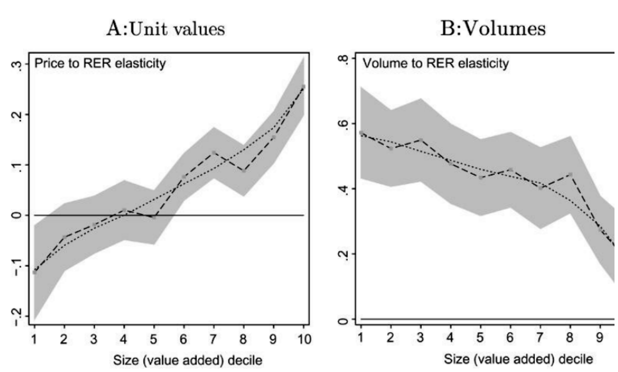


*Figure I: responses by size decile*

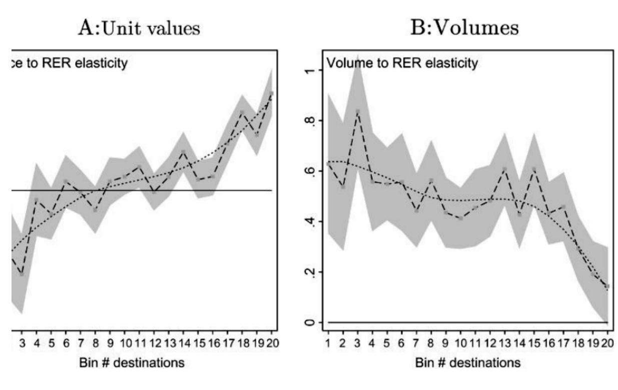


*Figure II: responses by destination-count bins*

## 7.5 Table VI: alternative mechanisms

**Read:**

- Table VI tests whether alternative channels explain the baseline patterns.
- The results are robust to product rank, import intensity, sector-level demand elasticities, and other controls.
- The negative volume interaction and positive price interaction survive these additional mechanisms.

**Economic message:** The heterogeneous price and volume responses are not fully explained by product ranking, import costs, or sector-specific demand differences.



TABLE VI
ALTERNATIVE MECHANISMS

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
|---|---|---|---|---|---|---|---|---|
| Dep. var. | Unit value | Export volume | Unit value | Export volume | Unit value | Export volume | Unit value | |
| # observations | 355996 | 355996 | 355874 | 355874 | 355996 | 355996 | 322397 | 289742 |
| ln TFP$_{t-1}$ | 0.012$^{a}$ | 0.083$^{a}$ | 0.013$^{a}$ | 0.082$^{a}$ | 0.014$^{a}$ | 0.083$^{a}$ | 0.013$^{a}$ | 0.012$^{a}$ |
| | (0.004) | (0.008) | (0.004) | (0.008) | (0.004) | (0.009) | (0.004) | (0.004) |
| ln RER | 0.071$^{a}$ | 0.425$^{a}$ | 0.300$^{a}$ | 0.022 | 0.279$^{a}$ | 0.235$^{a}$ | 0.120$^{a}$ | 0.122$^{a}$ |
| | (0.021) | (0.047) | (0.057) | (0.120) | (0.034) | (0.089) | (0.021) | (0.030) |
| ln TFP$_{t-1}\times$ ln RER | 0.049$^{a}$ | −0.109$^{a}$ | 0.047$^{a}$ | −0.106$^{a}$ | 0.033$^{b}$ | −0.106$^{a}$ | 0.046$^{a}$ | 0.048$^{a}$ |
| | (0.015) | (0.035) | (0.015) | (0.035) | (0.017) | (0.039) | (0.015) | (0.015) |
| $\frac{\text{imports}}{\text{sales}}\times$ ln RER | 0.136$^{c}$ | −0.283$^{c}$ | | | | | | |
| | (0.078) | (0.156) | | | | | | |
| BW06 $\sigma\times$ ln RER | | | −0.207$^{a}$ | 0.361$^{a}$ | | | | |
| | | | (0.048) | (0.102) | | | | |
| Mean $\Delta$ unit val. $\times$ ln RER | | | | | | | −0.063$^{b}$ | |
| | | | | | | | (0.029) | |
| sd. unit val. $\times$ ln RER | | | | | | | | −0.094 |
| | | | | | | | | (0.085) |
| ln GDP | | 0.627$^{a}$ | | 0.629$^{a}$ | | 0.628$^{a}$ | | |
| | | (0.051) | | (0.051) | | (0.051) | | |
| ln importer price index | | 0.054$^{a}$ | | 0.054$^{a}$ | | 0.054$^{a}$ | | |
| | | (0.012) | | (0.012) | | (0.012) | | |
| Sector $\times$ RER dummies | No | No | No | No | Yes | Yes | No | No |

*Note.* Robust standard errors clustered by destination-year in parentheses with a, b, and c, respectively, denoting significance at 1%, 5%, and 10%. All estimations include firm-destination fixed effects and year dummies. TFP is demeaned. BW06 $\sigma$ stands for the log of elasticity of substitution estimated in Broda and Weinstein (2006).

## 7.6 Tables VII, VIII, and IX: sector heterogeneity and distribution costs

**Read:**

- Table VII decomposes the productivity interaction into within-sector and between-sector components.
- Most of the heterogeneous response comes from within-sector heterogeneity among firms, not merely from sector composition.
- Table VIII tests the distribution-cost mechanism. The price and volume elasticities differ more strongly when distribution costs are high.
- Table IX compares intermediate and consumer goods. The distribution-cost channel is more relevant for consumer goods where local distribution margins matter more.

**Economic message:** The mechanism operates inside sectors and is consistent with the idea that local distribution costs amplify pricing-to-market by high-performance firms.

TABLE VII

WITHIN- VERSUS BETWEEN-SECTORS HETEROGENEITY

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | 114 sectors | | 700 sectors | |
| | Coef. | Contrib. | Coef. | Contrib. |
| Dep. var: ln unit values | | | | |
| ln RER | $0.084^{a}$ | | $0.085^{a}$ | |
| | (0.019) | | (0.019) | |
| ln $TFP_{t-1}\times$ ln RER (within) | $0.044^{a}$ | 73% | $0.046^{a}$ | 63% |
| | (0.016) | | (0.016) | |
| ln $TFP_{t-1}\times$ ln RER (between) | $0.044^{a}$ | 27% | $0.045^{a}$ | 37% |
| | (0.015) | | (0.015) | |
| Dep. var: ln export volume | | | | |
| ln RER | $0.398^{a}$ | | $0.397^{a}$ | |
| | (0.044) | | (0.044) | |
| ln $TFP_{t-1}\times$ ln RER (within) | $-0.098^{a}$ | 72% | $-0.092^{b}$ | 59% |
| | (0.038) | | (0.037) | |
| ln $TFP_{t-1}\times$ ln RER (between) | $-0.100^{a}$ | 28% | $-0.098^{a}$ | 41% |
| | (0.036) | | (0.036) | |

*Note*. Number of observations respectively 355,996 and 352,141 (this lower number is due to missing values in the more detailed definition of a sector). Robust standard errors clustered by destination-year in parentheses with $^{a}$, $^{b}$, and $^{c}$, respectively, denoting significance at 1%, 5%, and 10%. All estimations include

*Table VII: within- versus between-sector heterogeneity*

TABLE VIII

DISTRIBUTION COSTS

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| Distribution costs | High | Low | All | High | Low | High | Low |
| TFP | all | all | all | high | high | low | low |
| # observations | 50658 | 43730 | 94049 | 21852 | 25381 | 25374 | 21781 |
| Dep. var. | | | | ln unit value | | | |
| ln $TFP_{t-1}$ | $0.020^{b}$ | 0.012 | $0.018^{a}$ | $0.034^{c}$ | 0.030 | $0.036^{b}$ | $-0.016$ |
| | (0.008) | (0.010) | (0.006) | (0.018) | (0.023) | (0.014) | (0.013) |
| ln RER | $0.233^{a}$ | 0.065 | $0.320^{a}$ | $0.349^{a}$ | 0.064 | 0.112 | 0.059 |
| | (0.081) | (0.101) | (0.112) | (0.104) | (0.140) | (0.084) | (0.099) |
| Distrib. costs $\times$ ln RER | | | $0.113^{c}$ | | | | |
| | | | (0.063) | | | | |
| Difference in RER coeffs.‡ | 0.168 | | | | 0.290$^{b}$ | | |
| | (0.103) | | | | (0.122) | | |
| Dep. var. | | | | ln volume | | | |
| ln $TFP_{t-1}$ | $0.135^{a}$ | $0.100^{a}$ | $0.116^{a}$ | $0.123^{b}$ | $0.139^{a}$ | $0.071^{b}$ | $0.089^{a}$ |
| | (0.022) | (0.022) | (0.016) | (0.056) | (0.040) | (0.034) | (0.032) |
| ln RER | 0.036 | $0.530^{b}$ | $-0.075$ | $-0.363^{c}$ | 0.316 | 0.201 | $0.794^{a}$ |
| | (0.163) | (0.215) | (0.224) | (0.195) | (0.265) | (0.196) | (0.248) |
| Distrib. costs $\times$ ln RER | | | $-0.243^{c}$ | | | | |
| | | | (0.138) | | | | |
| ln GDP | $0.745^{a}$ | $0.704^{b}$ | $0.711^{a}$ | $1.020^{b}$ | $1.270^{a}$ | 0.316 | 0.235 |
| | (0.247) | (0.323) | (0.234) | (0.441) | (0.390) | (0.326) | (0.381) |
| ln importer price index | $0.244^{a}$ | 0.073 | $0.215^{a}$ | $0.288^{a}$ | $-0.159$ | 0.157 | 0.098 |
| | (0.076) | (0.085) | (0.066) | (0.088) | (0.120) | (0.102) | (0.124) |
| Difference in RER coeffs.‡ | $-0.494^{b}$ | | | | $-1.157^{a}$ | | |
| | (0.228) | | | | 0.270 | | |

*Note*. Robust standard errors clustered by destination-year in parentheses with $^{a}$, $^{b}$, and $^{c}$, respectively, denoting significance at 1%, 5%, and 10%. All estimations include firm-destination fixed effects and year dummies. Campa and Goldberg (2005) data for destination-sector specific distribution costs. ‡Difference in RER coefficients between columns (1) and (2), and between (4) and (7).

*Table VIII: distribution costs*

TABLE IX

ROBUSTNESS: INTERMEDIATES VERSUS CONSUMER GOODS

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Dep. var. | | ln unit value | | |
| Type of goods: | Interm. | Consum. | Interm. | Consum. |
| # observations | 152239 | 117321 | 152239 | 117321 |
| ln $TFP_{t-1}$ | $0.017^{a}$ | $0.023^{a}$ | $0.018^{a}$ | $0.013^{a}$ |
| | (0.005) | (0.005) | (0.006) | (0.005) |
| ln RER | $0.075^{a}$ | $0.199^{a}$ | $0.072^{a}$ | $0.180^{a}$ |
| | (0.023) | (0.023) | (0.023) | (0.024) |
| ln $TFP_{t-1}\times$ ln RER | | | $-0.017$ | $0.090^{a}$ |
| | | | (0.024) | (0.017) |
| Difference in ln RER coeffs. | $-0.124^{a}$ | | $-0.108^{a}$ | |
| | (0.026) | | (0.027) | |
| Difference in ln $TFP_{t-1}\times$ ln RER coeffs. | | | $-0.107^{a}$ | |
| | | | (0.029) | |

*Note*. Robust standard errors clustered by destination-year in parentheses with $^{a}$, $^{b}$, and $^{c}$, respectively,

## 7.7 Table X: extensive margin

**Read:**

- Table X studies the probability that a firm exports to a destination.
- Both TFP and real exchange-rate depreciations raise export participation.
- A 10% depreciation increases the probability of exporting by roughly 1.8 percentage points in the full sample.
- The effect is present both for entry into markets and for the probability of continuing to export.

**Economic message:** Exchange-rate movements affect not only prices and quantities conditional on exporting, but also the extensive margin of market participation.

TABLE X
THE EXTENSIVE MARGIN

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| Dep. var. | | | | | Probability of exporting | | | | |
| Condition | none | $x_{t-1}=0$ | $x_{t-1}=1$ | none | $x_{t-1}=0$ | $x_{t-1}=1$ | none | $x_{t-1}=0$ | $x_{t-1}=1$ |
| ln TFP$_{t-1}$ | $0.024^{a}$ | $0.016^{a}$ | $0.024^{a}$ | $0.024^{a}$ | $0.016^{a}$ | $0.024^{a}$ | $0.032^{a}$ | $0.020^{a}$ | $0.039^{a}$ |
| | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.001) | (0.002) |
| ln RER | $0.188^{a}$ | $0.139^{a}$ | $0.136^{a}$ | $0.184^{a}$ | $0.136^{a}$ | $0.138^{a}$ | $0.192^{a}$ | $0.153^{a}$ | $0.211^{a}$ |
| | (0.018) | (0.013) | (0.027) | (0.018) | (0.013) | (0.028) | (0.018) | (0.019) | (0.047) |
| ln GDP | $0.192^{a}$ | $0.146^{a}$ | $0.124^{a}$ | $0.178^{a}$ | $0.140^{a}$ | $0.127^{a}$ | $0.194^{a}$ | $0.149^{a}$ | $0.203^{a}$ |
| | (0.021) | (0.015) | (0.018) | (0.020) | (0.015) | (0.019) | (0.021) | (0.022) | (0.041) |
| ln importer | $0.024^{a}$ | $0.020^{a}$ | $0.021^{a}$ | $0.024^{a}$ | $0.020^{a}$ | $0.022^{a}$ | $0.025^{a}$ | $0.019^{a}$ | $0.022^{b}$ |
| price index | (0.007) | (0.005) | (0.005) | (0.006) | (0.005) | (0.005) | (0.001) | (0.006) | (0.010) |
| Observations | 2213558 | 1475999 | 737559 | 2213558 | 1475999 | 737559 | 2213558 | 1475999 | 737559 |
| Estimation | | Logit | | | LPM | | | LPM | |
| Fixed effects | | Destination | | | Destination | | | Firm-Dest. | |

*Note.* Robust standard errors clustered by destination-year in parentheses with $^{a}$, $^{b}$, and $^{c}$, respectively, denoting significance at 1%, 5%, and 10%. Columns (1)–(6) include destination fixed effects and year dummies, while columns (7)–(9) include firm-destination fixed effects and year dummies. The logit columns report average marginal effects.

## 7.8 Tables XI and XII: aggregate implications

**Read:**

- Table XI shows that sectors with more heterogeneity or concentration have lower exchange-rate elasticities of aggregate export values and volumes.
- The Herfindahl interaction and Pareto-shape interaction support the prediction that concentration among productive firms dampens aggregate responses.
- Table XII uses the model to decompose aggregate export elasticities into intensive and extensive components.
- The model can generate lower aggregate elasticities when high-performance firms dominate exports and absorb exchange-rate changes in markups.

**Economic message:** The firm-level heterogeneity documented earlier has macro consequences: it helps explain why aggregate exports respond weakly to exchange-rate movements.

TABLE XI
SECTOR-LEVEL IMPLICATIONS

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Dep. var. | ln sector-destination exports (total) | | (continuing exporters) | |
| ln RER | $0.916^{a}$ | $0.810^{a}$ | $0.940^{a}$ | $0.821^{a}$ |
| | (0.090) | (0.081) | (0.092) | (0.082) |
| Herfindahl index | $1.728^{b}$ | | $1.808^{b}$ | |
| | (0.824) | | (0.890) | |
| Herfindahl index × ln RER | $-0.559^{a}$ | | $-0.590^{a}$ | |
| | (0.175) | | (0.189) | |
| $k$ | | $-1.903^{b}$ | | $-1.630^{c}$ |
| | | (0.827) | | (0.862) |
| $k$ × ln RER | | $0.460^{a}$ | | $0.420^{b}$ |
| | | (0.175) | | (0.182) |
| ln GDP | $0.929^{a}$ | $1.026^{a}$ | $0.929^{a}$ | $1.032^{a}$ |
| | (0.078) | (0.085) | (0.084) | (0.091) |
| ln importer price index | $0.104^{a}$ | $0.120^{a}$ | $0.090^{a}$ | $0.108^{a}$ |
| | (0.020) | (0.022) | (0.022) | (0.024) |

*Table XI: sector-level implications*

TABLE XII
CALIBRATION OF AGGREGATE EXPORT ELASTICITIES TO EXCHANGE RATE

| | Data | | Model | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Data | | Benchmark | | $k=4$ | | $\sigma=4$ | | $\sigma=10$ | | $s_i=0.5$ | |
| | Vol. | Val. | Vol. | Val. | Vol. | Val. | Vol. | Val. | Vol. | Val. | Vol. | Val. |
| Intensive | 0.87 | 1.17 | 1.21 | 1.89 | 2.89 | 3.27 | 0.75 | 1.36 | 1.41 | 2.24 | 0.73 | 1.36 |
| Extensive | 0.08 | 0.10 | 0.29 | 0.61 | 1.11 | 1.73 | 0.75 | 1.14 | 0.09 | 0.56 | 0.77 | 1.14 |
| Total | 0.95 | 1.27 | 1.5 | 2.5 | 4 | 5 | 1.5 | 2.5 | 1.5 | 2.5 | 1.5 | 2.5 |

*Note.* This table provides an evaluation of equation (6), where the benchmark values of the parameters

*Table XII: aggregate export elasticity calibration*

## 7.9 Appendix robustness: Table A.1

**Read:**

- Table A.1 repeats sector-level tests with alternative sector-level productivity measures.
- The results are consistent with Table XI: sectors with more high-performance exporters have lower aggregate export elasticities.

**Economic message:** The aggregate implication is robust to alternative definitions of sector-level performance heterogeneity.

### *B. Robustness Results*

TABLE A.1

ROBUSTNESS: SECTOR LEVEL

| Dep.Var. | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | ln sector-destination exports (total) | | ln sector-destination exports (continuing exporters) | |
| ln RER | 1.168[a] | 1.034[a] | 1.188[a] | 1.020[a] |
| | (0.150) | (0.138) | (0.144) | (0.135) |
| mean sect. ln $\text{TFP}_{t-1}$ | $-0.006$ . | | 0.015 | |
| | (0.030) | | (0.031) | |
| mean sect. ln $\text{TFP}_{t-1} \times$ ln RER | $-0.341$[a] | | $-0.345$[a] | |
| | (0.118) | | (0.115) | |
| median sect. ln $\text{TFP}_{t-1}$ | | $-0.100$[a] | | $-0.121$[a] |
| | | (0.034) | | (0.035) |
| median sect. ln $\text{TFP}_{t-1} \times$ ln RER | | $-0.276$[b] | | $-0.247$[b] |
| | | (0.123) | | (0.122) |
| ln GDP | 1.032[a] | 1.029[a] | 1.039[a] | 1.035[a] |
| | (0.084) | (0.085) | (0.091) | (0.091) |
| ln importer price index | 0.121[a] | 0.121[a] | 0.109[a] | 0.109[a] |
| | (0.022) | (0.022) | (0.024) | (0.024) |

# 8 Short summary

- The paper studies how French exporters respond to real exchange-rate movements using firm-destination-product customs data from 1995–2005.
- High-performance firms raise export prices more after a depreciation, meaning they absorb more exchange-rate changes in markups.
- High-performance firms increase export volumes less after a depreciation because foreign-currency prices fall less for them.
- The result is robust to different samples, productivity measures, product-rank controls, distribution-cost controls, and sector-level decompositions.
- Depreciations also raise export participation probabilities, both through entry and continuation.
- The aggregate export response is dampened because exports are concentrated among the high-performance firms with the lowest volume elasticities.
- The paper provides micro evidence for variable-markup and distribution-cost models of incomplete exchange-rate pass-through.

# 9 Conclusion

## 9.1 Core conclusion

Berman, Martin, and Mayer show that exchange-rate pass-through and export-volume responses are systematically heterogeneous across exporters. High-performance exporters react to a depreciation primarily by raising markups and prices, whereas lower-performance exporters adjust quantities more strongly. This heterogeneity helps explain why aggregate exports may be relatively insensitive to exchange-rate changes.

## 9.2 Economic intuition

The key intuition is that productive exporters have lower perceived demand elasticities and greater market power. When the euro depreciates, these firms can keep much of the gain as higher markups rather than expanding volume aggressively. Since large firms account for a disproportionate share of exports, the aggregate export elasticity is dominated by precisely the firms that respond least in quantities.

## 9.3 Takeaway

The paper's main lesson is that exchange-rate elasticities are not structural constants shared by all exporters. They depend on firm performance, market power, distribution costs, and exporter composition. Any macro model of trade elasticities or pass-through that ignores firm heterogeneity can misstate both micro responses and aggregate export dynamics.